Target validation paths — hgsoc-tuboovarian-stage3c-r0-hrd-pending-idyq

The diagnostic and biomarker workup that hardens the targetable-feature call

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Target validation paths

Every biomarker in this case has resulted, but two workups are still worth ordering before acting on the tumor's most contingent options. A validated genomic-instability score decides whether olaparib plus bevacizumab (PAOLA-1, NCT02477644) is available: a Myriad MyChoice CDx GIS of at least 42 opens that combination, and below 42 the tumor reads HRD-negative and it is foreclosed. A confirmatory HER2 IHC re-stain decides whether a HER2-directed ADC such as trastuzumab deruxtecan is on the table: the IHC 1+ call rests on a single specimen Altera could not stain, and a re-stain reading HER2 0 takes the option away. Neither workup disturbs the all-comer PARP, anti-angiogenic, endocrine, or on-study choices. The remaining rows below record assays that have already resulted, kept here so the workup picture is complete.

HRD / genomic instability

The Altera tumor NGS calls genomic instability qualitatively, reporting chromosomal imbalances and frequent LOH, but it gives no validated quantitative score, and a qualitative call does not meet the companion-diagnostic definition of HRD-positive. This matters because the PAOLA-1 olaparib-plus-bevacizumab indication defines HRD-positive as either a deleterious BRCA mutation or a Myriad MyChoice CDx GIS of at least 42. The tumor is BRCA-wild-type, so eligibility rests entirely on the GIS. This is the one essential, decision-gating workup left: it runs on archival FFPE, with the surgical resection block preferred for tumor content, turnaround 2 to 3 weeks. The MyChoice CDx is the only assay whose threshold maps directly to the olaparib-plus-bevacizumab label. FoundationOne CDx and Caris report LOH-based HRD surrogates, so confirm the score maps to the at-least-42 threshold before relying on either for eligibility. Germline 74-gene panel and tumor BRCA1/2 sequencing have both resulted as wild-type, which forecloses BRCA-gated olaparib monotherapy (SOLO1) and routes the PARP decision onto this score; no further BRCA testing is outstanding.

HER2

The tumor scored HER2 IHC 1+ on a single surgical-pathology stain, which Libby reads as HER2-low. That call gates whether a HER2-directed ADC such as trastuzumab deruxtecan is worth discussing, because the tumor-agnostic trastuzumab-deruxtecan approval keys on IHC 3+. At IHC 1+ in ovarian cancer the drug is trial-eligibility and cross-tumor extrapolation only, never on-label. The 0-versus-1+ boundary has the poorest interrater concordance of any HER2 tier, and the Altera platform could not evaluate HER2 at all for lack of material, so no orthogonal read currently backs the stain. Two steps come first and are both high priority. Check specimen sufficiency on the surgical resection block (macrodissect if tumor content is low) before ordering anything, since an exhausted block forces a fresh-biopsy decision. Then run a confirmatory re-stain on the FDA-approved Ventana Pathway HER2 (clone 4B5) assay, scored to ASCO-CAP criteria, and ask for a discrete-tier report (0 / ultralow / 1+ / 2+ with reflex ISH / 3+) rather than a binary positive or negative call, because the tiers open different option sets: IHC 3+ or 2+/ISH-amplified puts the tumor-agnostic label in range, HER2-low keeps it at trial-eligibility, and HER2-ultralow is a still-lower tier defined only in breast. Reflex HER2 ISH/FISH (ERBB2 dual-probe) is conditional, triggered only if the re-stain reads 2+; do not order it up front on the current 1+. ERBB2 copy-number on a comprehensive NGS panel is a low-priority orthogonal cross-check, since HER2-low protein is expression-driven and a normal copy number neither confirms nor refutes the 1+ IHC; protein IHC stays the decision-driving assay. The IHC work runs on archival FFPE, turnaround 1 to 2 weeks, low cost.

ER

ER is positive in roughly 80 percent of nuclei at weak-to-moderate intensity with PR negative, which supports a low-toxicity aromatase-inhibitor option such as letrozole. PR-negativity predicts a more modest endocrine response, so the semiquantitative ER/PR readout sets expectations rather than gating a drug. The values are already reported; no additional staining is needed unless a later specimen is obtained.

PIK3CA / MAP2K4

The Altera panel reported focal PIK3CA amplification and focal MAP2K4 deletion, both investigational-only in HGSC with no approved matched therapy. They matter only as potential early-phase trial-matching tags on the PI3K/AKT/mTOR axis or MAP2K4-listed studies. No confirmatory or orthogonal assay is warranted; the findings are recorded for trial-screening completeness.

Signatera ctDNA (MRD)

Tumor-informed ctDNA has cleared to Not Detected after cytoreduction and chemoimmunotherapy, marking a favorable molecular-residual-disease state that informs how intensively to pursue maintenance. Serial monitoring can flag biochemical relapse ahead of CA-125 or imaging. It does not select a target or gate a drug, so this is surveillance rather than a one-time gating assay; continuity favors keeping draws on the platform that designed the patient-specific panel.

Where to order these assays

Assay	Provider	Decision gated	Contact
Validated genomic-instability score (Myriad MyChoice CDx GIS, or FDA-approved equivalent companion diagnostic)	**Myriad Genetics (preferred) (MyChoice CDx)**	**Olaparib plus bevacizumab maintenance (PAOLA-1) eligibility in BRCA-wild-type HGSC, which is gated on a validated GIS of at least 42.**	**test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423**
Validated genomic-instability score (Myriad MyChoice CDx GIS, or FDA-approved equivalent companion diagnostic)	Foundation Medicine (FoundationOne CDx (LOH-based HRD))	Olaparib plus bevacizumab maintenance (PAOLA-1) eligibility in BRCA-wild-type HGSC, which is gated on a validated GIS of at least 42.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Validated genomic-instability score (Myriad MyChoice CDx GIS, or FDA-approved equivalent companion diagnostic)	Caris Life Sciences (Caris HRD / genomic LOH)	Olaparib plus bevacizumab maintenance (PAOLA-1) eligibility in BRCA-wild-type HGSC, which is gated on a validated GIS of at least 42.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
HER2 IHC re-stain (Ventana Pathway HER2 clone 4B5), scored to ASCO-CAP criteria	**NeoGenomics Laboratories (preferred) (HER2 (4B5) IHC)**	**Whether a HER2-directed ADC (e.g. trastuzumab deruxtecan) is on the table; tumor-agnostic T-DXd needs IHC 3+, so a confirmed HER2-low 1+ is trial-eligibility only in ovarian cancer.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**

HER2 IHC re-stain (Ventana Pathway HER2 clone 4B5), scored to ASCO-CAP criteria	Labcorp Oncology (HER2 IHC (4B5))	Whether a HER2-directed ADC (e.g. trastuzumab deruxtecan) is on the table; tumor-agnostic T-DXd needs IHC 3+, so a confirmed HER2-low 1+ is trial-eligibility only in ovarian cancer.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
HER2 IHC re-stain (Ventana Pathway HER2 clone 4B5), scored to ASCO-CAP criteria	Quest Diagnostics (HER2 IHC)	Whether a HER2-directed ADC (e.g. trastuzumab deruxtecan) is on the table; tumor-agnostic T-DXd needs IHC 3+, so a confirmed HER2-low 1+ is trial-eligibility only in ovarian cancer.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
HER2 IHC re-stain (Ventana Pathway HER2 clone 4B5), scored to ASCO-CAP criteria	Caris Life Sciences (Caris MI Profile (IHC + WTS))	Whether a HER2-directed ADC (e.g. trastuzumab deruxtecan) is on the table; tumor-agnostic T-DXd needs IHC 3+, so a confirmed HER2-low 1+ is trial-eligibility only in ovarian cancer.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
HER2 IHC re-stain (Ventana Pathway HER2 clone 4B5), scored to ASCO-CAP criteria	Tempus (Tempus xT / IHC)	Whether a HER2-directed ADC (e.g. trastuzumab deruxtecan) is on the table; tumor-agnostic T-DXd needs IHC 3+, so a confirmed HER2-low 1+ is trial-eligibility only in ovarian cancer.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137
Specimen-sufficiency / tissue-adequacy check on the surgical resection block	**NeoGenomics Laboratories (preferred) (Tissue QC / macrodissection)**	**Enables the confirmatory HER2 IHC, reflex ISH, and ERBB2 NGS reads; gates whether any HER2-directed ADC workup can proceed.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
Specimen-sufficiency / tissue-adequacy check on the surgical resection block	Caris Life Sciences (Tissue QC / molecular profiling)	Enables the confirmatory HER2 IHC, reflex ISH, and ERBB2 NGS reads; gates whether any HER2-directed ADC workup can proceed.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
Full HER2-tier classification (3+ vs 2+/ISH vs HER2-low vs HER2-ultralow)	**NeoGenomics Laboratories (preferred) (HER2 (4B5) IHC, full-tier reporting)**	**Which HER2 stratum applies: 3+ (on-label tumor-agnostic T-DXd), HER2-low (trial-eligibility ADC), or ultralow; each opens a different option set.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
Full HER2-tier classification (3+ vs 2+/ISH vs HER2-low vs HER2-ultralow)	Labcorp Oncology (HER2 IHC (4B5))	Which HER2 stratum applies: 3+ (on-label tumor-agnostic T-DXd), HER2-low (trial-eligibility ADC), or ultralow; each opens a different option set.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167

Full HER2-tier classification (3+ vs 2+/ISH vs HER2-low vs HER2-ultralow)	Quest Diagnostics (HER2 IHC)	Which HER2 stratum applies: 3+ (on-label tumor-agnostic T-DXd), HER2-low (trial-eligibility ADC), or ultralow; each opens a different option set.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Reflex HER2 ISH/FISH (ERBB2 dual-probe), triggered only on a 2+ stain	**NeoGenomics Laboratories (preferred) (HER2 (ERBB2) FISH)**	**Separates HER2-low (2+/ISH-negative) from HER2-amplified (2+/ISH-positive); conditional on a 2+ re-stain.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
Reflex HER2 ISH/FISH (ERBB2 dual-probe), triggered only on a 2+ stain	Labcorp Oncology (HER2 FISH)	Separates HER2-low (2+/ISH-negative) from HER2-amplified (2+/ISH-positive); conditional on a 2+ re-stain.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
Reflex HER2 ISH/FISH (ERBB2 dual-probe), triggered only on a 2+ stain	Quest Diagnostics (HER2 FISH)	Separates HER2-low (2+/ISH-negative) from HER2-amplified (2+/ISH-positive); conditional on a 2+ re-stain.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
ERBB2 copy-number / expression on a comprehensive NGS panel	**Foundation Medicine (preferred) (FoundationOne CDx (ERBB2 copy number))**	**Orthogonal ERBB2 amplification check; does not gate ADC eligibility, which is set by protein IHC.**	**test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639**
ERBB2 copy-number / expression on a comprehensive NGS panel	Tempus (Tempus xT (ERBB2 copy number))	Orthogonal ERBB2 amplification check; does not gate ADC eligibility, which is set by protein IHC.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137
ERBB2 copy-number / expression on a comprehensive NGS panel	Caris Life Sciences (Caris MI Profile (ERBB2 copy number))	Orthogonal ERBB2 amplification check; does not gate ADC eligibility, which is set by protein IHC.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669
Serial tumor-informed ctDNA MRD (Signatera, already cleared to Not Detected)	**Natera (preferred) (Signatera)**	**Maintenance intensity and relapse surveillance; does not gate a specific drug.**	**test info · 13011 McCallen Pass, Building A, Austin, TX 78753 · 1-650-980-9190**

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements

Validated genomic-instability score (Myriad MyChoice CDx GIS, or FDA-approved equivalent companion diagnostic)	The Altera tumor NGS calls genomic instability qualitatively (chromosomal imbalances and frequent LOH) but reports no validated quantitative score, and a qualitative call does not satisfy the companion-diagnostic definition of HRD-positive. The PAOLA-1 maintenance indication requires HRD-positive status defined as a deleterious BRCA mutation or a Myriad MyChoice CDx GIS of at least 42; this tumor is BRCA wild-type, so eligibility rests entirely on the GIS. Without a validated score the patient defaults to all-comer niraparib maintenance (PRIMA) and the higher-benefit option in HRD-positive disease cannot be offered.	Myriad Genetics (MyChoice CDx) · test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423	archival FFPE acceptable (surgical resection block preferred for tumor content)
Germline 74-gene panel plus tumor BRCA1/2 sequencing (already performed)	Both germline (74-gene panel) and somatic (Altera tumor/normal exome) BRCA1/2 testing have resulted as wild-type, which is the resolution that previously gated the case. This forecloses the BRCA-mutation-gated indications (olaparib monotherapy maintenance, SOLO1) and shifts the PARP question onto the genomic-instability score. No further BRCA testing is needed; the row records the resolved state so downstream agents do not re-flag it as pending.	Resolved on germline blood and tumor block already tested; no additional provider order required	none additional; germline blood and tumor block already tested

ER and PR semiquantitative IHC (Allred or percentage plus intensity, already reported)	ER is positive in roughly 80 percent of nuclei at weak-to-moderate intensity with PR negative, which supports a low-toxicity aromatase-inhibitor option (for example letrozole) as maintenance or later-line therapy that fits the patient's age and oral-preference. PR-negativity predicts a more modest endocrine response, so the quantitative ER/PR readout sets expectations rather than gating a drug. The values are already reported; no additional staining is required unless a later specimen is obtained.	Reported on the diagnostic IHC panel; no additional provider order required	archival FFPE acceptable
Comprehensive tumor NGS reporting PIK3CA amplification and MAP2K4 deletion (already performed, Altera)	The Altera panel reported focal PIK3CA amplification and focal MAP2K4 deletion, both investigational-only in HGSC with no approved matched therapy. They are relevant only as potential early-phase trial-matching tags (PI3K/AKT/mTOR-axis or MAP2K4-listed studies). No confirmatory or orthogonal assay is warranted; the findings are recorded for trial-screening completeness.	Reported on the Altera tumor NGS panel; no additional provider order required	archival FFPE acceptable

Serial tumor-informed ctDNA MRD (Signatera, already cleared to Not Detected)	Signatera tumor-informed ctDNA has cleared to Not Detected after cytoreduction and chemoimmunotherapy, which marks a favorable state and informs how intensively to pursue maintenance. Serial monitoring can detect biochemical relapse ahead of CA-125 or imaging and help time maintenance decisions. It does not select a target or gate any specific drug, so priority is medium and surveillance, not a one-time gating assay.	Natera (Signatera) · test info · 13011 McCallen Pass, Building A, Austin, TX 78753 · 1-650-980-9190	serial whole-blood draws; tumor block already used for assay design
HER2 IHC re-stain (Ventana Pathway HER2 clone 4B5), scored to ASCO-CAP criteria	The 1+ call rests on a single surgical-pathology stain, and the 0-versus-1+ boundary has the poorest interrater concordance of any HER2 IHC tier. A confirmatory re-stain on the FDA-approved Ventana Pathway HER2 (4B5) assay, scored against ASCO-CAP criteria, hardens whether the tumor is genuinely HER2-low; the surgical report used ASCO-CAP gastric / DESTINY-PanTumor scoring, which labels 1+ as 'negative' even though the case-level tier is HER2-low. Skipping it leaves any HER2-directed ADC discussion resting on an unconfirmed, low-concordance score.	NeoGenomics Laboratories (HER2 (4B5) IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE acceptable (additional unstained slides from the surgical resection block)

/ tissue-adequacy check on the surgical resection block	The Altera tumor NGS could not evaluate HER2 because of insufficient material, so no orthogonal molecular read currently backs the single IHC. Confirm the surgical resection block has enough residual tumor for added unstained sections (macrodissect if tumor content is low) before ordering the 4B5 re-stain and any ERBB2 copy-number read; if the block is exhausted, weigh a fresh biopsy against the modest yield of repeat HER2 testing. Without adequate tissue the HER2-low call cannot be confirmed or extended to ISH or NGS.	NeoGenomics Laboratories (Tissue QC / macrodissection) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE block (assess residual tumor; fresh biopsy only if block exhausted)
Full HER2-tier classification (3+ vs 2+/ISH vs HER2-low vs HER2-ultralow)	A binary positive/negative HER2 call does not tell the board which option applies, because the tiers diverge sharply. IHC 3+ (or 2+/ISH-amplified) would put the tumor-agnostic trastuzumab-deruxtecan label in range; HER2-low (1+ or 2+/ISH-negative) keeps T-DXd at trial-eligibility and cross-tumor extrapolation from breast (DESTINY-Breast04); HER2-ultralow (IHC 0 with faint incomplete staining, per DESTINY-Breast06) is a lower tier defined only in breast. Reporting the discrete tier rather than a label is what lets the trial_screener match the correct HER2 stratum.	NeoGenomics Laboratories (HER2 (4B5) IHC, full-tier reporting) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE acceptable (same sections as the confirmatory stain)

Reflex HER2 ISH/FISH (ERBB2 dual-probe), triggered only on a 2+ stain	Reflex ISH/FISH is indicated only when a stain scores 2+, to separate HER2-low (2+/ISH-negative) from gene-amplified disease (2+/ISH-positive). The current 1+ does not trigger ISH, so this row is conditional on the confirmatory re-stain coming back 2+. The tumor-agnostic trastuzumab-deruxtecan label keys on IHC 3+ rather than ISH in non-breast tumors, so amplification status here informs the biological call and trial stratification more than a direct drug gate.	NeoGenomics Laboratories (HER2 (ERBB2) FISH) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE acceptable (reflex on the same block; no new specimen)
ERBB2 copy-number / expression on a comprehensive NGS panel	ERBB2 copy number and expression on a comprehensive NGS panel gives the orthogonal molecular read the Altera run could not produce on this tumor. In HER2-low disease the protein is driven by expression rather than gene amplification, so a normal ERBB2 copy number neither confirms nor refutes the 1+ IHC; an unexpected amplification would prompt slide re-review. Protein IHC stays the decision-driving assay for ADC eligibility, so this is a low-priority cross-check, not a substitute.	Foundation Medicine (FoundationOne CDx (ERBB2 copy number)) · test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639	archival FFPE acceptable (depends on the specimen-sufficiency check clearing)

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.